Assessment 1 - QnA Section

Islington College,Nepal

Document Details

Submission ID

trn:oid:::3618:138383554

Submission Date

May 10, 2026, 5:59 PM GMT+5:45

Download Date

May 10, 2026, 6:36 PM GMT+5:45

File Name

2408972_NishantJaiswal_qna.docx

File Size

65.3 KB

4 Pages

1,310 Words

6,981 Characters

14% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Filtered from the Report

- Small Matches (less than 8 words)

Exclusions

- 1 Excluded Source
- 12 Excluded Matches

Match Groups

16 Not Cited or Quoted 14%
Matches with neither in-text citation nor quotation marks

0 Missing Quotations 0%
Matches that are still very similar to source material

0 Missing Citation 0%
Matches that have quotation marks, but no in-text citation

0 Cited and Quoted 0%
Matches with in-text citation present, but no quotation marks

Top Sources

3%	Internet sources
1%	Publications
14%	Submitted works (Student Papers)

Match Groups

- **16** Not Cited or Quoted 14%
  Matches with neither in-text citation nor quotation marks
- **0** Missing Quotations 0%
  Matches that are still very similar to source material
- **0** Missing Citation 0%
  Matches that have quotation marks, but no in-text citation
- **0** Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

Top Sources

3% Internet sources
1% Publications
14% Submitted works (Student Papers)

Top Sources

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

1 Submitted works
Islington College,Nepal on 2026-05-10 **9%**

2 Internet
www.mdpi.com **1%**

3 Submitted works
Manchester Metropolitan University on 2024-01-11 **1%**

4 Submitted works
Universiti Kebangsaan Malaysia on 2011-08-20 **<1%**

5 Submitted works
Islington College,Nepal on 2026-05-10 **<1%**

6 Submitted works
Islington College,Nepal on 2026-05-10 **<1%**

7 Submitted works
Islington College,Nepal on 2026-05-10 **<1%**

HERALD COLLEGE
KATHMANDU

UNIVERSITY OF WOLVERHAMPTON

2026	Part-I Question and Answer	1	Report

Part-I Question and Answer

Github Link : https://github.com/Uknowme-h/AIML_CourseWork

Student Id : 2408972

Student Name : Nishant Jaiswal

Section : L6CG4

Module Leader : Mr. Siman Giri

Tutor : Mr. Anish Khatiwada

Long Question — Answer 1 of 2 [5 Marks]

Question 1: You are a Data Scientist at eSewa, a leading digital payment platform, working on large-scale user transaction data. Identify two high-impact business problems where unsupervised learning can provide value. For each problem, define it in context, propose a suitable approach, discuss system integration, and briefly mention one limitation.

Problem 1: Account Takeover Detection via Session Behaviour Analysis

Business Context : As a digital wallet, eSewa is one of the prime targets of attackers. An attacker can achieve full access to a victim's account by using malware, session hijacking or phishing. The actual issue isn't actually the login itself it's the actions the attacker takes after logging in.

Proposed Approach:

LSTM Autoencoder: The user sessions are always sequential. Legitimate user actions are orderly, rhythmic and purposeful. A normal autoencoder reduces a session to a vector and discards all temporal information. LSTM cells solve this problem: the input gate decides what to add to the state of the cell, the forget gate determines what to forget and the output gate decides what to keep. The model learns the transitions as well as the actions. A sequence such as 'balance check , small transfer , logout' is well reconstructed. A sequence such as 'login, immediate maximum transfer, add new beneficiary, logout' has a high reconstruction error since those transitions did not occur in any of the training legitimate sessions. Only normal sessions are used for training the model. An attacker that is walking on paths he does not know or skipping steps that a real user would not skip, will incur reconstruction error at inference without the attacker being warned, which re-verifies his OTP in the middle of the session. Navigation sequences, time per screen, input cadence, and transaction speed are features used.

System Integration : Any number of sessions flow through RabbitMQ (or any other message broker). Scoring is not just at login. If the anomaly score exceeds a threshold, then silent OTP re-verification is done before any fund movement is made.

Limitation : Users on new devices or in new environments produce "true" false positives, which requires per-user threshold calibration, not a single global threshold.

Problem 2: API Abuse and Enumeration Attack Detection

Business Context : eSewa's APIs for account verification, merchant lookup and balance checking are targeted to map valid accounts and/or to get merchant info. The slow and sub-rate-limited requests are intentionally designed in this way so that they will not be detected by threshhold detection.

Proposed Approach: LSTM Autoencoder on API Call Sequences : There is structure in legitimate API usage: calls follow logical sequences based on true user intent: authenticate, fetch balance, initiate payment, confirm. The LSTM Autoencoder is used to learn this temporal grammar. It represents a session's API call sequence as a latent vector, or behavioural intent, via an encoder. The decoder is used to recreate the original sequence. With legal sessions, the reconstruction error is low. Structurally empty: Enumeration attacks are attacks in which the same endpoints are called with incrementally different parameters, but the flow is not logical. Such repetitive patterns are never useful in the LSTM's hidden state; otherwise, it is the forget gate's job to forget them. Reconstruction error remains high throughout, which is an indication of a legitimate session error that usually drops as going along the expected progression of the sequence. Features: end-point identifiers, inter-call intervals, parameter variation from call-to-call, and response code patterns.

System Integration: A 5 minute run session window, and a near real-time batch job. Throttling occurs without interruption (silently) - responses slow down but do not block completely, attackers don't know they are detected.

Limitation : Enumeration has a consistent API structure to third party integration and partner merchant reconciliation jobs. If these are not upstream whitelisted for known automated clients, they will continue to create an exponentially increasing number of false positives.

Short Questions Pick 1 from Each Category [5 Marks]

Category 1: Overfitting [2.5 Marks]

Question 2: Overfitting is a common challenge in deep learning, especially when models have high capacity. Describe at least two techniques used to reduce overfitting, explaining the intuition behind each, how it affects training, and a real-world application example.

Technique 1: Dropout

When a neural network has more capacity than the training data it is trained on, it memorises the training data rather than learning general patterns that is overfitting. This is prevented by dropout, which randomly inhibits a portion of the neurons for each training step. The network can never trust any single neuron, and thus has to learn distributed, robust representations. During the test phase every neuron is re-enabled, however, the output of each of these neurons is multiplied in order to account for dropout in the training phase. In the case of image classification for KYC validation, dropout helps the model focus on more than a few details from a single area of the image; it forces the network to learn from several different cues in an image rather than just one or two.

Technique 2: Data Augmentation

In some cases, a model may be poorly designed but it is also because of the small size of the training data or lack of diversity in the training data. Data augmentation increases the effective size of the training set by creating variations of existing data using transformations such as flipping, rotating, cropping, changing brightness or noise to existing examples, without acquiring any new data. The model is now presented with slightly modified versions of each of the examples, and now the task of memorising is difficult while the task of generalising is easy. Add rotations, brightness variations and blurriness to a cheque or document scanner when they are only used with a small number of examples, and real-world performance is significantly boosted.

Category 2: Neural Network Architecture [2.5 Marks]

Question 2: FCN vs. Autoencoder Compare a standard feedforward neural network (FCN) with an autoencoder in terms of architecture, objective, and input-output relationship. Provide an example and describe one application of autoencoders.

Architecture

A fully connected network is one where a flow of information goes one way from the input layer to the hidden layers and from the hidden layers to the output layer, with each neuron in one layer connected to each neuron in another layer. The structure is the same as an autoencoder, but with a “bottleneck”: An autoencoder compresses the input into a smaller latent representation, then reconstructs the original input from that compressed form.

Objective and Functionality

Supervised learning is used by an FCN to map inputs to the predefined labels. An autoencoder is unsupervised it has no external labels. It only needs to learn the most crucial structure in the data, as its only goal is to reconstruct its own input as accurately as possible.

Input-Output Relationship

The concept of input and output in an FCN is quite different: an image is given as an input, a class label is returned as an output. In an autoencoder, the input is the same as the output: an image is fed into the system and an approximated/denoised version of the same image is generated.

Example and Application

Consider input vector [0.9, 0.1, 0.8, 0.95]. An FCN outputs a class prediction ('cat' or 'dog'). It is compressed and denoised by an Autoencoder to yield denoised output, such as [0.98, 0.02, 0.97, 0.99]. A real KYC pipeline would train an autoencoder using only legal citizenship documents to capture their structure and patterns at the pixel level. The decoder fails to correctly decode the document if it is forged when it passes through the system; the latent vector does not contain the unknown patterns in the forgery. Without any labelled fake documents, this reconstruction error is an outlier and warns of the anomalous nature of the document.